

Daniel Benson
Anderson Morris

Task 1:

The first task was to simply train a PCA model on our training data of the survivors. The goal was to reduce the dimensionality while maintaining at least 90% of the data's original variance. To do this, we used the recommended PCA library from scikit-learn which allowed us to specify that we wanted our model to only keep components that add up to the desired 90% variance, then fit our model onto our training data. Finally, we then saved our model using the pickle library, and saved our weights after transforming our data with the PCA model.

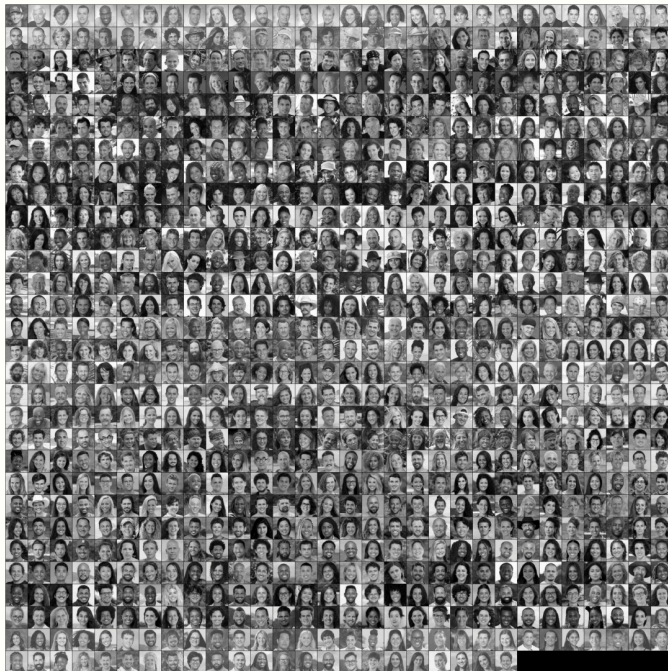


Figure 1: Montage photo of training dataset

Task 2:

In order to determine which professor looks least like a face, we had to reconstruct each professor's face into what's called an eigenface. These are made using the eigenvector components that were kept in our 90% variance from earlier, creating "ghost-like" images of each professor. Finally, we project these faces into a 3D "face space" to then compare their weights with their original image weights. Whichever professor's eigenface has the greatest value (largest distance) in comparison with their original photo is then determined to look the least like a face. Ultimately, it was found that Dr. Roberson looked least like a face, with a total euclidean distance of 194.869.



Figure 2: Each professor's eigenface compared with their original photo

Task 3:

The next question asked to predict which professor is most likely to host the next season of Survivor. To do this, we must first project each professor's reconstructed image into the "face space", then apply the nearest-neighbors method to determine who is closest (looks most similar) to the host, Jeff Probst. The professors were reconstructed using the same method in problem 2, however the nearest-neighbors algorithm was implemented using the respective scikit-learn library. A model was trained on 5 nearest neighbors (one for each professor) and fitted to the test data of the professors. In the end, the model predicted using k-nearest neighbors, that Dr. Eicholtz would host the next season of Survivor, due to looking the most similar to Jeff Probst (who would've thought).



Figure 3: Reconstructed faces of Jeff Probst and the next host, Dr. Eicholtz

Task 4:

To predict which season of Survivor each professor is most likely to be on, we used k-means clustering with our trained PCA model, and assigned each professor to the nearest cluster. Finally, we find the mode in each cluster, which corresponds to the season that the majority of survivors contested in, and assign each professor to the season of the cluster they were matched with. To explain in more detail, the scikit-learn library for the k-means algorithm was used on the weights from the survivors. Then, the professors are projected into the face-space and assigned a cluster by fitting them to the k-means model. Finally, for each professor, we took the mode Survivor season for all nodes in their cluster, and assigned the resulting season to the respective professor. This resulted in:

- Dr. Roberson → Season 38
- Dr. Cazalas → Season 1
- Dr. Burke → Season 20
- Dr. Eicholtz → Season 7
- Dr Touraj → Season 26

Task 5:

The final task for the project was to determine which professor is most likely to win Survivor. Furthermore, we were tasked with coming up with our own quantitative metric to predict who might win. Initially, we thought about using the results from the past 3 problems to help us find a prospective candidate, but then realized that the answers from the previous tasks did not relate to a professor winning. In the end, we decided to simply compare each of the professors to the winners across all of the seasons. In order to accomplish this, we loaded and filtered the .csv file of show rankings to just track the first place winners. Then, we simply loaded our survivor weights, PCA model and test data, and began iterating over our survivors. If any of the survivors in our training data matched with a name from the first-place rankings list, we would use the index they were found at to retrieve their weights, and append it to an array. This process was repeated for each survivor until we got the weights from each winner. After converting the array of weights to a numpy array, we calculated the mean, then measured the euclidean distance from each professor to the mean average of the winner's mean (just like in problem 2, but searching for the opposite result!). The script returned at the end that the predicted winner of the next season of Survivor would be Dr. Eicholtz (shocker.).



Figure 4: Mean winner of Survivor vs predicted winner eigenface and original image